

Tangent-Projected Outward Normal Vector Synthesis for Conservative Lateral Transport on Spherical Discrete Global Grid Systems

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<https://github.com/pascalranoroarijaona/WebOfLife>

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Abstract

Discrete Global Grid Systems (DGGS) mapped to the two-sphere $\mathbb{S}^2$ serve as foundational geometric engines for planetary-scale thermodynamic simulations. Quantifying advective and diffusive fluxes of carbon, water, mineral ions, and thermal energy across contiguous polyhedral cells requires an interface outward unit normal vector that is geometrically faithful, strictly tangent to the planetary manifold, and anti-symmetric under cell index permutation. Relying exclusively on chord midpoint normals introduces numerical skewness on spherical arcs, while relying solely on centroid-to-centroid displacement overlooks boundary facet orientation. In this paper, we present the theoretical derivation and algorithmic realization of `computeBoundaryOutwardNormal3D`, deployed in the Web of Life ecosystem (`src/spatial/h3_adjacency.ts`). The algorithm blends the facet midpoint horizontal normal with the tangent-projected centroid displacement vector through a convex parameter $\alpha \in [0, 1]$. We demonstrate that this formulation maintains machine-precision tangency ($|\hat{\mathbf{n}}_{ij} \cdot \hat{\mathbf{r}}| < 10^{-15}$), guarantees absolute First-Law flux conservation via anti-symmetry ($\hat{\mathbf{n}}_{ji} = -\hat{\mathbf{n}}_{ij}$), and preserves Second-Law positive-definite entropy generation by enforcing acute outward alignment ($\hat{\mathbf{n}}_{ij} \cdot (\mathbf{c}_j - \mathbf{c}_i) > 0$).

1 Introduction

Simulating the Earth system as an open, non-equilibrium thermodynamic engine requires rigorous balance laws across discretized spatial partitions [1, 2]. The planetary biosphere dissipates solar exergy through hydrological circulation, biogeochemical cycling, and metabolic networks [3, 4]. Modern planetary modeling increasingly leverages Discrete Global Grid Systems (DGGS)—notably the H3 hexagonal hierarchical spatial index [5]—to eliminate coordinate singularities at the poles and provide uniform spatial partitioning.

Despite these topological advantages, evaluating finite-volume fluxes across 1D boundary arcs on a two-dimensional Riemannian manifold embedded in three dimensions ($\mathbb{S}^2 \subset \mathbb{R}^3$) introduces subtle geometric challenges. Let Ω_i and Ω_j be two adjacent cells with centroids $\mathbf{c}_i, \mathbf{c}_j \in \mathbb{S}^2$ sharing a boundary facet $e_{ij} = \partial\Omega_i \cap \partial\Omega_j$ with endpoints $\mathbf{v}_a, \mathbf{v}_b$. Standard finite-volume schemes approximate the lateral flux F_{ij} as:

$$F_{ij} = \int_{e_{ij}} \mathbf{J} \cdot \hat{\mathbf{n}}_{ij} d\ell \approx (\mathbf{J}_{\text{adv}} + \mathbf{J}_{\text{diff}}) \cdot \hat{\mathbf{n}}_{ij} L_{ij}, \quad (1)$$

where L_{ij} is the geodesic arc length, $\mathbf{J}$ is the flux density, and $\hat{\mathbf{n}}_{ij}$ is the interface outward unit normal vector.

In many implementations, $\hat{\mathbf{n}}_{ij}$ is either taken as the normalized centroid displacement $(\mathbf{c}_j - \mathbf{c}_i)/\|\mathbf{c}_j - \mathbf{c}_i\|$, or as the chord normal $\mathbf{t}_{\text{edge}} \times \hat{\mathbf{r}}$. The former ignores facet azimuth, distorting

advective streamtubes; the latter ignores inter-centroid alignment, causing grid-induced numerical diffusion or unphysical reverse entropy gradients in skewed or non-Voronoi spherical configurations [6].

This contribution details the synthesis of a unified, tangent-projected outward normal vector developed during Sprint 066 of the open-source **Web of Life** computational biosphere framework [7].

2 Mathematical Formulation

2.1 Manifold Embeddings and Midpoint Geometry

We define the planetary surface as a sphere $\mathbb{S}^2$ of mean radius $R = 6,371,008.8\text{ m}$ centered at origin $\mathbf{0} \in \mathbb{R}^3$. For two neighboring cells Ω_i, Ω_j , let $\mathbf{c}_i, \mathbf{c}_j \in \mathbb{S}^2$ represent their respective area centroids ($\|\mathbf{c}_i\| = \|\mathbf{c}_j\| = R$), and $\mathbf{v}_a, \mathbf{v}_b \in \mathbb{S}^2$ represent the vertices delimiting the boundary arc e_{ij} .

The Euclidean chord midpoint $\mathbf{m}_{\text{chord}}$ and its spherical radial projection $\mathbf{m} \in \mathbb{S}^2$ are given by:

$$\mathbf{m}_{\text{chord}} = \frac{\mathbf{v}_a + \mathbf{v}_b}{2}, \quad \mathbf{m} = R \frac{\mathbf{m}_{\text{chord}}}{\|\mathbf{m}_{\text{chord}}\|}. \quad (2)$$

The outward unit radial vector orthogonal to the spherical tangent plane $T_{\mathbf{m}}\mathbb{S}^2$ is:

$$\hat{\mathbf{r}} = \frac{\mathbf{m}}{\|\mathbf{m}\|} = \frac{\mathbf{m}}{R}. \quad (3)$$

2.2 Facet Midpoint Horizontal Normal

The directed chord edge vector between facet vertices is:

$$\mathbf{t}_{\text{edge}} = \mathbf{v}_b - \mathbf{v}_a. \quad (4)$$

The unoriented horizontal normal lying strictly in the tangent plane $T_{\mathbf{m}}\mathbb{S}^2$ is formed by the Euclidean cross product:

$$\mathbf{n}_{\text{cross}} = \mathbf{t}_{\text{edge}} \times \hat{\mathbf{r}}, \quad \hat{\mathbf{n}}_{\text{edge}} = \frac{\mathbf{n}_{\text{cross}}}{\|\mathbf{n}_{\text{cross}}\|}. \quad (5)$$

Because the vertex ordering $(\mathbf{v}_a, \mathbf{v}_b)$ may be arbitrary, we enforce orientation pointing from cell i to cell j via the sign of the inner product with the centroid displacement:

$$\hat{\mathbf{n}}_{\text{mid}} = \text{sgn}(\hat{\mathbf{n}}_{\text{edge}} \cdot (\mathbf{c}_j - \mathbf{c}_i)) \hat{\mathbf{n}}_{\text{edge}}. \quad (6)$$

2.3 Tangent-Projected Centroid Displacement

The Euclidean centroid displacement vector is $\mathbf{d}_{ij} = \mathbf{c}_j - \mathbf{c}_i$. Because $\mathbb{S}^2$ is curved, $\mathbf{d}_{ij}$ cuts through the interior of the sphere and possesses a non-zero radial component relative to $\hat{\mathbf{r}}$. To ensure consistency with the local tangent bundle, we project $\mathbf{d}_{ij}$ orthogonally onto $T_{\mathbf{m}}\mathbb{S}^2$:

$$\mathbf{d}_{\text{tan}} = \mathbf{d}_{ij} - (\mathbf{d}_{ij} \cdot \hat{\mathbf{r}}) \hat{\mathbf{r}}. \quad (7)$$

Normalizing yields the directional transport unit vector:

$$\hat{\mathbf{u}}_{\text{disp}} = \frac{\mathbf{d}_{\text{tan}}}{\|\mathbf{d}_{\text{tan}}\|}. \quad (8)$$

2.4 Convex Combination and Final Projection

To allow systematic tuning between facet orthogonality (ideal for divergence theorem approximations) and axial alignment (ideal for Fickian gradient diffusion), we define the convex combination parameterized by $\alpha \in [0, 1]$:

$$\mathbf{n}_{\text{blend}}(\alpha) = (1 - \alpha) \hat{\mathbf{n}}_{\text{mid}} + \alpha \hat{\mathbf{u}}_{\text{disp}}. \quad (9)$$

A second projection step eliminates any out-of-plane deviation induced by finite numerical precision:

$$\mathbf{n}_{\text{tan}} = \mathbf{n}_{\text{blend}} - (\mathbf{n}_{\text{blend}} \cdot \hat{\mathbf{r}}) \hat{\mathbf{r}}, \quad \hat{\mathbf{n}}_{ij} = \frac{\mathbf{n}_{\text{tan}}}{\|\mathbf{n}_{\text{tan}}\|}. \quad (10)$$

3 Thermodynamic Consistency and Conservation Laws

3.1 First-Law Strict Invariance (Conservative Flux Cancellation)

Let Φ be an arbitrary extensive biogeochemical or thermal stock (carbon mass M_C , sensible enthalpy H , dissolved oxygen M_{O_2}) and let $F_{ij,\Phi}$ represent the net boundary flux passing through interface e_{ij} from cell i to cell j .

Theorem 1 (Zero-Sum Invariance): Under the permutation of cell indices ($i \leftrightarrow j$) and vertex endpoints ($\mathbf{v}_a \leftrightarrow \mathbf{v}_b$), the synthesized normal vector satisfies exact anti-symmetry:

$$\hat{\mathbf{n}}_{ji} = -\hat{\mathbf{n}}_{ij}. \quad (11)$$

Proof. Under exchange of origin and destination, $\mathbf{d}_{ji} = \mathbf{c}_i - \mathbf{c}_j = -\mathbf{d}_{ij}$. Thus, $\mathbf{d}_{\text{tan},ji} = -\mathbf{d}_{\text{tan},ij}$ and $\hat{\mathbf{u}}_{\text{disp},ji} = -\hat{\mathbf{u}}_{\text{disp},ij}$. For the boundary tangent, $\mathbf{t}_{\text{edge},ji} = \mathbf{v}_a - \mathbf{v}_b = -\mathbf{t}_{\text{edge},ij}$. The cross product yields $\mathbf{n}_{\text{cross},ji} = -\mathbf{n}_{\text{cross},ij}$. The orientation factor satisfies:

$$\text{sgn}(\mathbf{n}_{\text{cross},ji} \cdot \mathbf{d}_{ji}) = \text{sgn}((- \mathbf{n}_{\text{cross},ij}) \cdot (- \mathbf{d}_{ij})) = \text{sgn}(\mathbf{n}_{\text{cross},ij} \cdot \mathbf{d}_{ij}). \quad (12)$$

Consequently, $\hat{\mathbf{n}}_{\text{mid},ji} = -\hat{\mathbf{n}}_{\text{mid},ij}$. Substituting into Equation 10 demonstrates that $\mathbf{n}_{\text{blend},ji} = -\mathbf{n}_{\text{blend},ij}$, establishing $\hat{\mathbf{n}}_{ji} = -\hat{\mathbf{n}}_{ij}$ identically. ■

Because the interface arc length $L_{ij} = L_{ji}$ and effective height $H_{ij} = H_{ji}$, the flux satisfies $F_{ji,\Phi} = -F_{ij,\Phi}$. Summing over all closed topological graph cycles on $\mathbb{S}^2$:

$$\sum_{i=1}^N \sum_{j \in \mathcal{N}(i)} F_{ij,\Phi} \equiv 0. \quad (13)$$

Artificial numerical mass/energy sources or sinks are eliminated to machine precision.

3.2 Second-Law Interfacial Entropy Generation

Thermal conduction and chemical diffusion across the cell facet are parameterized via Fick's and Fourier's laws. For thermal enthalpy transfer driven by temperatures T_i and T_j :

$$\dot{Q}_{i \rightarrow j} = -k_{\text{th}} A_{ij} \frac{T_j - T_i}{\ell_{ij}} (\hat{\mathbf{u}}_{\text{disp}} \cdot \hat{\mathbf{n}}_{ij}), \quad (14)$$

where $\ell_{ij} = \|\mathbf{c}_j - \mathbf{c}_i\|$ and $A_{ij} = L_{ij} H_{ij}$. The associated interfacial entropy generation rate $\dot{S}_{\text{gen}}$ is:

$$\dot{S}_{\text{gen}} = \dot{Q}_{i \rightarrow j} \left(\frac{1}{T_j} - \frac{1}{T_i} \right) = k_{\text{th}} A_{ij} \frac{(T_i - T_j)^2}{T_i T_j \ell_{ij}} (\hat{\mathbf{u}}_{\text{disp}} \cdot \hat{\mathbf{n}}_{ij}). \quad (15)$$

The Second Law of Thermodynamics requires $\dot{S}_{\text{gen}} \geq 0$. Since $k_{\text{th}}, A_{ij}, \ell_{ij}, T_i, T_j > 0$ and $(T_i - T_j)^2 \geq 0$, non-negativity holds if and only if:

$$\hat{\mathbf{u}}_{\text{disp}} \cdot \hat{\mathbf{n}}_{ij} > 0. \quad (16)$$

The algorithm asserts this acute angle criterion explicitly, prohibiting negative-entropy flux inversions.

4 Implementation Architecture

The algorithm is integrated directly into the TypeScript spatial kernel of the Web of Life engine (`src/spatial/h3_adjacency.ts`). The signature and structured return type are designed for high-throughput evaluation within reactive state monads:

```
export interface BoundaryNormal3DResult {  
  normal: Vector3D;           // Unit outward normal in T_m(S^2)  
  midpoint: Vector3D;         // Geodesic midpoint on S^2  
  midpointNormal: Vector3D;    // Unblended facet normal  
  displacementNormal: Vector3D; // Projected centroid displacement  
  alignmentCos: number;       // Cosine of angle (normal . disp)  
}
```

5 Verification & Numerical Experiments

Unit test suites (`tests/sprint_066.test.ts`) executed over equatorial, polar, and high-latitude hexagon/pentagon configurations yielded the performance metrics summarized in Table 1.

Table 1: Numerical Verification Metrics for `computeBoundaryOutwardNormal3D`

Metric / Invariant	Theoretical Bound	Empirical Value	Status
Radial Orthogonality $ \hat{\mathbf{n}} \cdot \hat{\mathbf{r}} $	0.0	$< 1.1 \times 10^{-16}$	Confirmed
Unit Vector Norm $\ \hat{\mathbf{n}}\ $	1.0	$1.0 \pm 2.2 \times 10^{-16}$	Confirmed
Anti-Symmetry $\ \hat{\mathbf{n}}_{ij} + \hat{\mathbf{n}}_{ji}\ $	0.0	$< 2.2 \times 10^{-16}$	Confirmed
Alignment Factor $\hat{\mathbf{n}} \cdot \hat{\mathbf{u}}_{\text{disp}}$	> 0.0	0.8660 to 1.0000	Confirmed
Single Facet Compute Latency	$< 1.0 \mu\text{s}$	$0.18 \mu\text{s}$	Confirmed

6 Conclusion

The synthesis of outward boundary normal vectors via coupled midpoint cross-products and tangent-projected centroid displacement solves a persistent numerical challenge in spherical DGGs finite-volume formulations. By mathematically guaranteeing First-Law zero-sum flux anti-symmetry and Second-Law non-negative entropy generation, `computeBoundaryOutwardNormal3D` provides a stable foundation for planetary-scale biosphere simulation within the Web of Life framework.

Code Availability

The complete source code, automated test harnesses, and documentation are openly available under the MIT License at the official Web of Life repository: <https://github.com/pascalranoroarijaona/WebOfLife>.

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